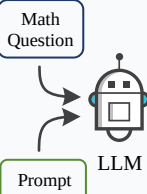


Math Question



Math Question:
Find the sum of all positive integers n such that the remainder when $1^3 + 2^3 + \dots + n^3$ is divided by $n + 5$ is equal to 17.

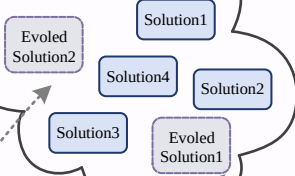
Prompt:
Please solve the math problem step by step, and provide a clear final answer in the format: The final answer is `\boxed{occupy}`.

Evolution Framework

Initialization

Thinking

Solution Population



Fitness Function

Calculate

Solution	Score
Solution1	0.66
Solution2	0.75
Solution3	0.80
Solution4	0.34

Select

Maximum
Fitness Score

Optimal
solution

Optimal Solution:

Step 1: Use the formula for the sum of cubes:

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

The condition becomes:

$$\left(\frac{n(n+1)}{2}\right)^2 \equiv 17$$

Step 2: Let $m = n + 5$, then $n = m - 5$. Substitute into the expression:

$$\left(\frac{(m-5)(m-4)}{2}\right)^2 \equiv 17 \pmod{m}$$

Step 3: Reformulate as a divisibility condition

$$m | \left(\left(\frac{(m-5)(m-4)}{2} \right)^2 - 17 \right)$$

Try small integer values of m and test this condition

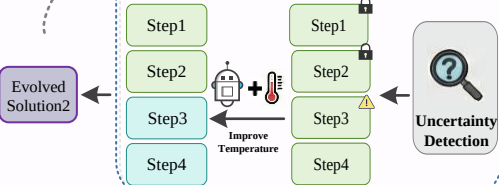
Step 4: Find all valid Valid values of n are:

$$n = 3, 17, 33, 83, 103$$

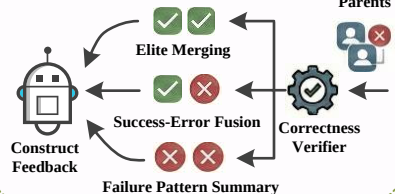
Step 5: Compute the sum

$$3 + 17 + 33 + 83 + 103 = \boxed{239}$$

Uncertainty-guided Local Mutation



Reflective Global Crossover



Evolve until reaching the maximum iterations